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REACTING MASS CALCULATIONS

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The first step in solving a problem is always to identify the information and quantities you are given and the ones you are required to calculate.

In Stoichiometry, you should first check the equation is BALANCED as part of determining what information you have been given.

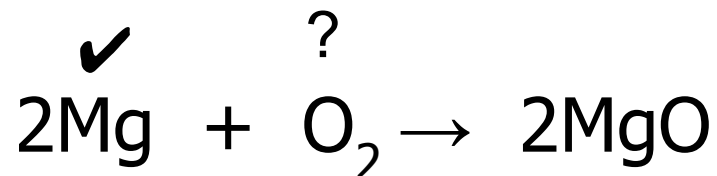
SYMBOLS USED
in these answers:

Given quantity ✓

Required Quantity ?

REACTING MASS CALCULATIONS (Examples)

- 1) What mass of oxygen reacts with 12 g of magnesium?



$$\text{Moles Mg} = \frac{\text{mass}}{\text{MM}} = \frac{12\text{g}}{24\text{g/mol}} = 0.5$$

$$\text{Moles O}_2 = 0.25 \text{ mol}$$

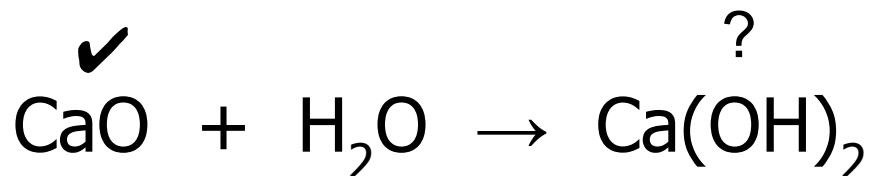
$$\begin{aligned}\text{Mass O}_2 &= \text{MM} \times \text{moles} \\ &= 32\text{g/mol} \times 0.25\text{mol} \\ &= \underline{8\text{g}}\end{aligned}$$

REACTING MASS CALCULATIONS

1. Write ✓ ? on balanced equation
2. Find moles ✓ $(= \frac{\text{mass}}{\text{MM}})$
3. Find moles ? (use balanced equation)
4. Find mass ? $(= \text{mass} \times \text{MM})$

REACTING MASS CALCULATIONS (Examples)

2) What mass of calcium hydroxide is made from 14 kg of calcium oxide?



NOTICE how units cancel out to leave you with the desired unit. This is a good mental check!!!

$$\text{Moles CaO} = \frac{\text{mass}}{\text{MM}} = \frac{14000 \text{ g}}{56\text{g/mol}} = 250 \text{ mol}$$

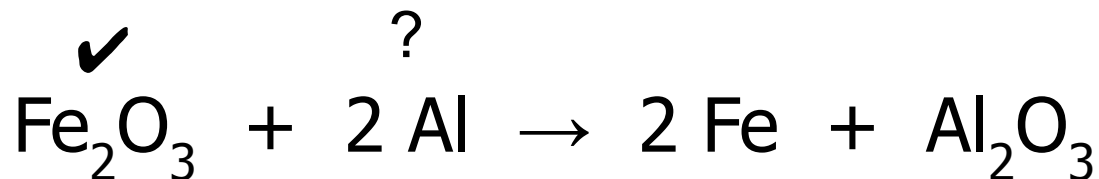
$$\text{Moles Ca(OH)}_2 = 250\text{mol}$$

$$\begin{aligned}\text{Mass Ca(OH)}_2 &= \text{MM} \times \text{moles} \\ &= 74\text{g/mol} \times 250\text{mol} \\ &= \underline{\underline{18500 \text{ g}}}\end{aligned}$$

It is an expectation that equations will be solved on multiple lines with the = signs lined up as per this equation. For fitting solutions on the slides some are solved horizontally, though

REACTING MASS CALCULATIONS (Examples)

3) What mass of aluminium is needed to react with 640 g of iron oxide?



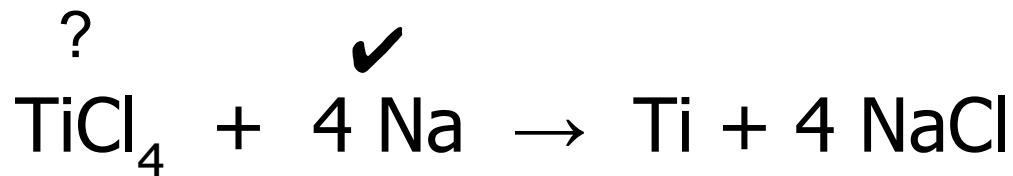
$$\text{Moles Fe}_2\text{O}_3 = \frac{\text{mass}}{\text{MM}} = \frac{640\text{g}}{160\text{g/mol}} = 4\text{mol}$$

$$\text{Moles Al} = n\text{Fe}_2\text{O}_3 \times \text{molar ratio} = 4\text{mol} \times \frac{2}{1} = 8\text{mol}$$

$$\text{Mass Al} = \text{MM} \times \text{moles} = 27\text{g/mol} \times 8\text{mol} = \underline{\underline{216 \text{ g}}}$$

REACTING MASS CALCULATIONS (Examples)

4) What mass of titanium chloride reacts with 460 g of sodium?



$$\text{Moles Na} = \frac{\text{mass}}{\text{MM}} = \frac{460}{23} = 20$$

$$\text{Moles TiCl}_4 = 5$$

$$\text{Mass TiCl}_4 = \text{MM} \times \text{moles} = 190 \times 5 = \underline{\underline{950 \text{ g}}}$$

The numbers and solutions are all correct on this slide. WHAT IS MISSING IN THE SOLUTIONS?? Hint: what should be after the numbers in each equation?